

Math 135 HW 4

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1. Let $u(x, t) = F(x + ct) + G(x - ct)$. Then

$$u_t = cF'(x + ct) - cG'(x - ct), \quad u_{tt} = c^2F''(x + ct) + c^2G''(x - ct),$$

$$u_x = F'(x + ct) + G'(x - ct), \quad u_{xx} = F''(x + ct) + G''(x - ct).$$

Hence $u_{tt} = c^2u_{xx}$.

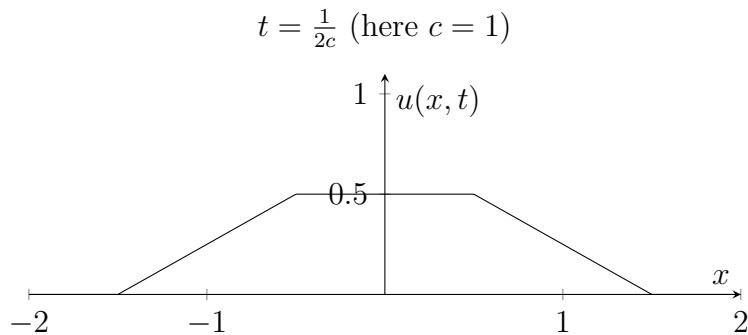
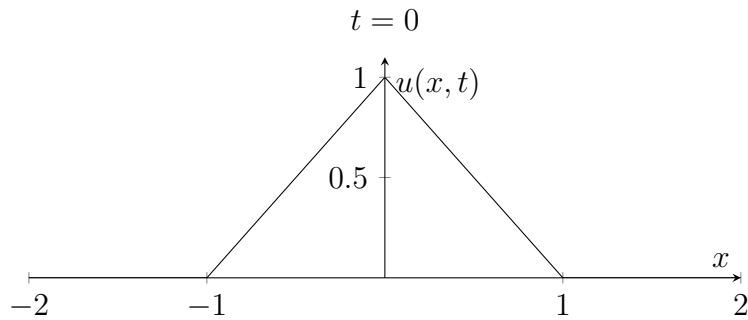
Given $u(x, 0) = f(x)$ and $u_t(x, 0) = 0$:

$$F(x) + G(x) = f(x), \quad cF'(x) - cG'(x) = 0 \Rightarrow F'(x) = G'(x).$$

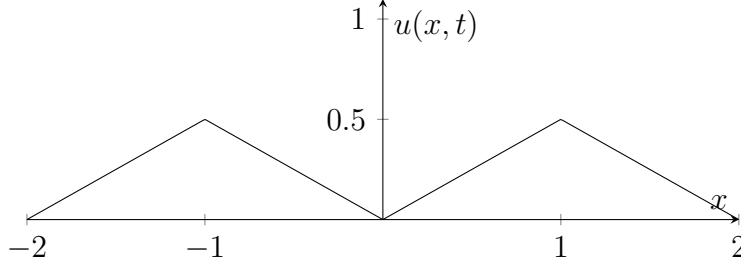
Thus $G(x) = F(x) + C$. Take $C = 0$. Then $F = G = \frac{1}{2}f$ and

$$u(x, t) = \frac{1}{2}[f(x + ct) + f(x - ct)].$$

For $f(x) = \max(1 - |x|, 0)$: two identical half height triangular pulses travel left and right with speed c .



$$t = \frac{1}{c} \text{ (here } c = 1\text{)}$$



2. Solve $X'' + \lambda X = 0$.

(a) $X'(0) = X'(L) = 0$. General $X = A \cos(\sqrt{\lambda}x) + B \sin(\sqrt{\lambda}x)$. From $X'(0) = 0$ get $B = 0$. Then $X'(L) = 0 \Rightarrow \sin(\sqrt{\lambda}L) = 0$.

$$\lambda_n = \left(\frac{n\pi}{L}\right)^2, \quad X_n(x) = \cos\left(\frac{n\pi x}{L}\right), \quad n = 0, 1, 2, \dots$$

(b) $X(0) = 0, X'(L) = 0$. From $X(0) = 0$ get $A = 0$. Then $X'(L) = 0 \Rightarrow \cos(\sqrt{\lambda}L) = 0$.

$$\lambda_n = \left(\frac{(2n-1)\pi}{2L}\right)^2, \quad X_n(x) = \sin\left(\frac{(2n-1)\pi x}{2L}\right).$$

(c) $X'(0) = 0, X(L) = 0$. From $X'(0) = 0$ get $B = 0$. Then $X(L) = 0 \Rightarrow \cos(\sqrt{\lambda}L) = 0$.

$$\lambda_n = \left(\frac{(2n-1)\pi}{2L}\right)^2, \quad X_n(x) = \cos\left(\frac{(2n-1)\pi x}{2L}\right).$$

3. $u_t = a^2 u_{xx}$ on $0 < x < L$, $u(0, t) = u(L, t) = 0$,

$$u(x, 0) = f(x) = \begin{cases} x, & 0 \leq x \leq L/2, \\ L - x, & L/2 < x \leq L. \end{cases}$$

(a) $\lambda_n = (n\pi/L)^2$, $X_n(x) = \sin(n\pi x/L)$.

(b) Fourier sine coefficients $C_n = \frac{2}{L} \int_0^L f(x) \sin\left(\frac{n\pi x}{L}\right) dx$. Evaluating,

$$C_n = \frac{4L}{n^2\pi^2} \sin\left(\frac{n\pi}{2}\right).$$

(c)

$$u(x, t) = \sum_{n=1}^{\infty} C_n e^{-a^2(n\pi/L)^2 t} \sin\left(\frac{n\pi x}{L}\right).$$

4. $u_t = u_{xx}$ on $0 < x < \pi$, $u(0, t) = 0$, $u(\pi, t) = 3$, $u(x, 0) = \frac{3x}{\pi} + \sin x$.

Steady state $v'' = 0$, $v(0) = 0$, $v(\pi) = 3 \Rightarrow v(x) = \frac{3x}{\pi}$. Let $w = u - v$. Then $w_t = w_{xx}$, $w(0, t) = w(\pi, t) = 0$, $w(x, 0) = \sin x$. Thus $w(x, t) = e^{-t} \sin x$, and

$$u(x, t) = \frac{3x}{\pi} + e^{-t} \sin x.$$

5. $u_t = a^2 u_{xx}$, $u_x(0, t) = u_x(L, t) = 0$, $u(x, 0) = \cos(\pi x/L)$. Neumann eigenfunctions $\cos(n\pi x/L)$. Only $n = 1$ appears:

$$u(x, t) = \cos\left(\frac{\pi x}{L}\right) \exp\left(-a^2 \left(\frac{\pi}{L}\right)^2 t\right).$$

The zero eigenmode $n = 0$ is absent since the initial data is not constant.

6. $u_t = u_{xx}$ on $0 < x < 1$, $u_x(0, t) = 0$, $u_x(1, t) + \alpha u(1, t) = 0$, $u(x, 0) = 1 - x$. Solve $X'' + \lambda X = 0$ with $X'(0) = 0$, $X'(1) + \alpha X(1) = 0$. With $X(x) = A \cos(kx)$ and $k = \sqrt{\lambda}$, the boundary at $x = 1$ gives

$$-k \sin k + \alpha \cos k = 0 \quad \Rightarrow \quad k \tan k = \alpha.$$

Let k_n be the positive roots. Then

$$u(x, t) = \sum_{n=1}^{\infty} C_n e^{-k_n^2 t} \cos(k_n x), \quad C_n = \frac{\int_0^1 (1-x) \cos(k_n x) dx}{\int_0^1 \cos^2(k_n x) dx}.$$

7. Wave equation $u_{tt} = c^2 u_{xx}$, $u(0, t) = u(L, t) = 0$, $u(x, 0) = \sin(3\pi x/L)$, $u_t(x, 0) = 0$.

$$u(x, t) = \sin\left(\frac{3\pi x}{L}\right) \cos\left(\frac{3\pi c t}{L}\right).$$

8. $u_{tt} = u_{xx}$ on $0 < x < \pi$, $u(0, t) = 0$, $u_x(\pi, t) = 0$,

$$u(x, 0) = \begin{cases} 0, & 0 \leq x < \pi/2, \\ 1, & \pi/2 \leq x \leq \pi, \end{cases} \quad u_t(x, 0) = 0.$$

Solve $X'' + \lambda X = 0$ with $X(0) = 0$, $X'(\pi) = 0$. This gives $X_n(x) = \sin((n - \frac{1}{2})x)$, $\lambda_n = (n - \frac{1}{2})^2$, $n = 1, 2, \dots$. Thus

$$u(x, t) = \sum_{n=1}^{\infty} A_n \sin((n - \frac{1}{2})x) \cos((n - \frac{1}{2})t),$$

$$A_n = \frac{2}{\pi} \int_0^{\pi} f(x) \sin((n - \frac{1}{2})x) dx, \quad f(x) = \begin{cases} 0, & 0 \leq x < \pi/2, \\ 1, & \pi/2 \leq x \leq \pi. \end{cases}$$

9. $u_{tt} = 4u_{xx}$ on $0 < x < \pi$, $u(0, t) = u(\pi, t) = 0$, $u(x, 0) = 0$, $u_t(x, 0) = 2 \sin(3x)$. Write $u(x, t) = \sum_{n \geq 1} (A_n \cos(2nt) + B_n \sin(2nt)) \sin(nx)$. Then $A_n = 0$ and $2nB_n = 2\delta_{n3} \Rightarrow B_3 = \frac{1}{3}$. Hence

$$u(x, t) = \frac{1}{3} \sin(3x) \sin(6t).$$

10. On $0 < x < a$, $0 < y < b$: $u_t = a_0^2(u_{xx} + u_{yy})$, $u = 0$ on the boundary, $u(x, y, 0) = \sin(\pi x/a) \sin(2\pi y/b)$.

$$u(x, y, t) = \sin\left(\frac{\pi x}{a}\right) \sin\left(\frac{2\pi y}{b}\right) \exp\left(-a_0^2 \left[\left(\frac{\pi}{a}\right)^2 + \left(\frac{2\pi}{b}\right)^2\right] t\right).$$

Decay rate in time is $a_0^2[(\pi/a)^2 + (2\pi/b)^2]$.

11. On $(0, \pi) \times (0, \pi)$: $u_t = u_{xx} + u_{yy}$, $u = 0$ on the boundary, $u(x, y, 0) = x \sin y$.

(a) Expand $x = \sum_{m=1}^{\infty} a_m \sin(mx)$ with

$$a_m = \frac{2}{\pi} \int_0^{\pi} x \sin(mx) dx = \frac{2(-1)^{m+1}}{m}.$$

So $u(x, y, 0) = \sum_{m=1}^{\infty} a_m \sin(mx) \sin(y)$.

(b) Series solution

$$u(x, y, t) = \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} A_{mn} \sin(mx) \sin(ny) e^{-(m^2+n^2)t},$$

with $A_{m1} = a_m$ and $A_{mn} = 0$ for $n \neq 1$. The slowest decaying present term is $m = 1, n = 1$.